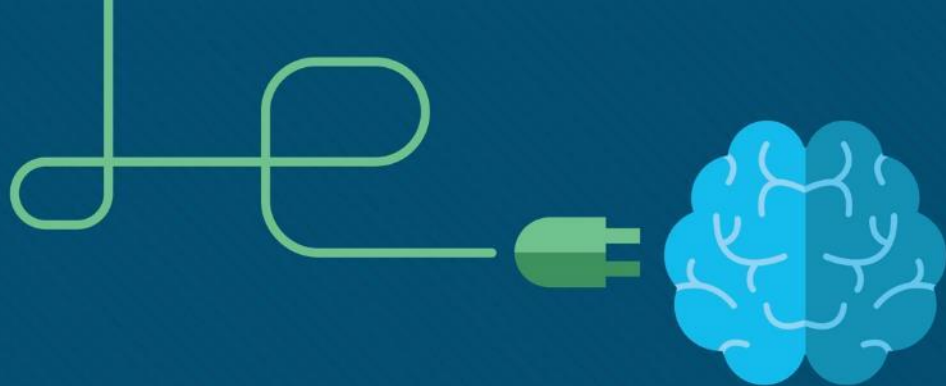




Chapter 10: Application Layer

Curriculum Title

Introduction to Networks v6.0



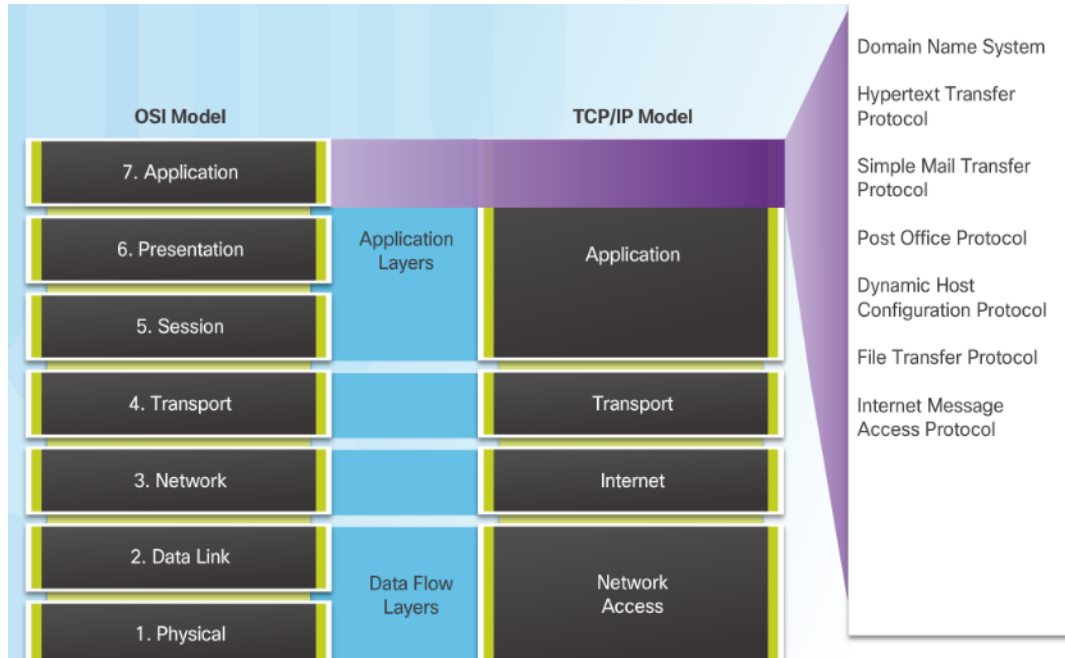
Chapter 10 - Sections & Objectives

- 10.1 Application Layer Protocols
 - Explain the operation of the application layer in providing support to end-user applications.
 - Explain how the functions of the application layer, session layer, and presentation layer work together to provide network services to end user applications
 - Explain how common application layer protocols interact with end user applications.
- 10.2 Well-Known Application Protocols and Services
 - Explain how well-known TCP/IP application layer protocols operate.
 - Explain how web and email protocols operate.
 - Explain how DNS and DHCP operate.
 - Explain how file transfer protocols operate.

10.1 Application Layer Protocols

Application, Presentation, and Session

Application Layer

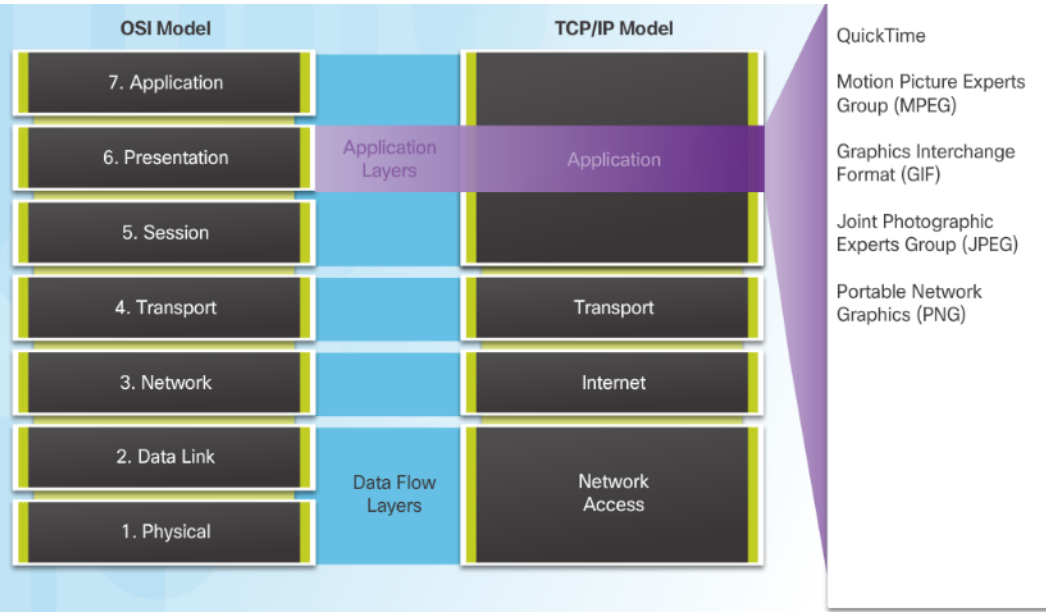


■ Application Layer:

- Closest to the end user.
- Used to exchange data between programs running on the source and destination hosts.

Application, Presentation, and Session

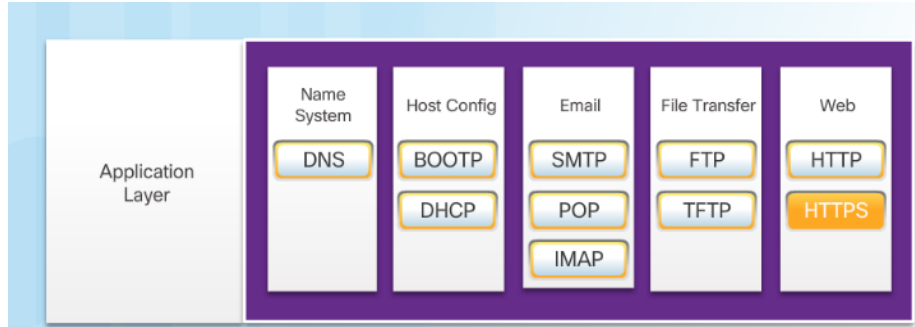
Presentation and Session Layer



- **Presentation Layer function:**
 - Formatting data at the source device into a compatible form for the receiving device.
 - Compressing data.
 - Encrypting data.
- **Session Layer Function**
 - Create and maintain dialogs between source and destination applications.

Application, Presentation, and Session

TCP/IP Application Layer Protocols

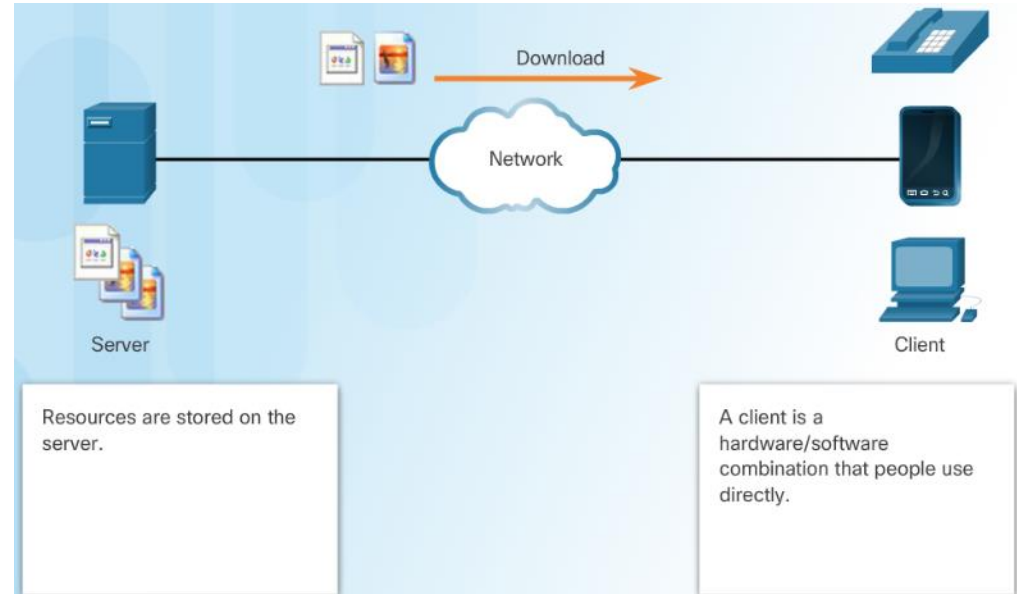


- Domain Name Server (DNS) TCP,UDP 53 - Translates domain names, such as cisco.com, into IP addresses.
- (BOOTP) – Bootstrap Protocol - BOOTP is being superseded by DHCP.
- Dynamic Host Configuration Protocol (DHCP) UDP client 68, server 67 – Dynamically assigns IP addresses to client stations at start-up.
- Simple Mail Transport Protocol (SMTP) TCP 25 - Enables clients to send email to a mail server.
- Post Office Protocol (POP) TCP 110 - Enables clients to retrieve email from a mail server.
- Internet Message Access Protocol (IMAP) TCP 143 - Enables clients to retrieve email from a mail server, maintains email on server.
- File Transfer Protocol (FTP) TCP 20 and 21 - Reliable, connection-oriented, and acknowledged file delivery protocol.
- Trivial File Transfer Protocol (TFTP) UDP 69 – simple connectionless file transfer protocol.
- Hypertext Transfer Protocol (HTTP) TCP 80, 8080 - Set of rules for exchanging text, graphic images, etc. on the World Wide Web.
- Hypertext Transfer Protocol Secure (HTTPS) TCP, UDP 443 – Uses encryption and authentication to secure communication.

How Application Protocols Interact with End-User Applications

Client-Server Model

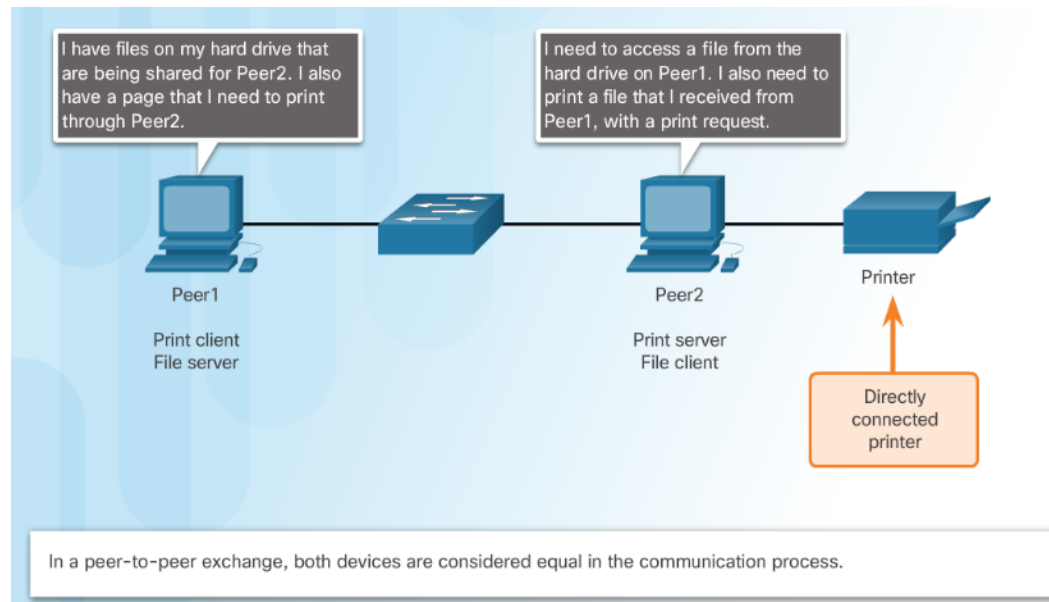
- Client and server processes are considered to be in the application layer.
- Application layer protocols describe the format of the requests and responses between clients and servers.
- Example of a client-server network is using an ISP's email service to send, receive and store email.



How Application Protocols Interact with End-User Applications

Peer-to-Peer Networks

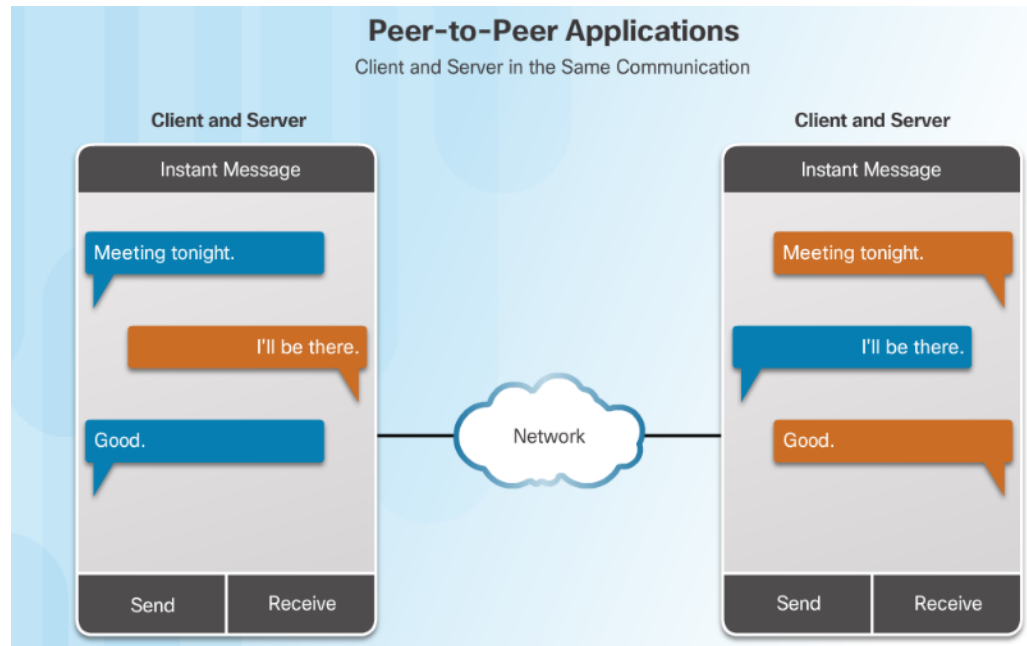
- Data is accessed from a peer device without the use of a dedicated server.
- Each device (known as a peer) can function as both a server and a client.



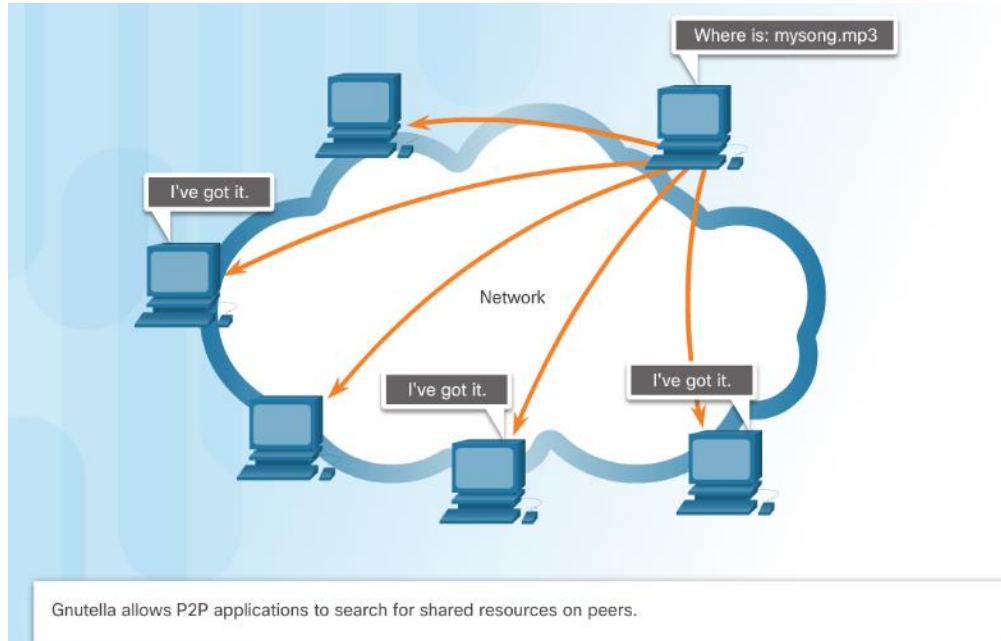
How Application Protocols Interact with End-User Applications

Peer-to-Peer Applications

- A P2P application allows a device to act as both a client and a server within the same communication.
- P2P applications require that each end device provide a user interface and run a background service.



Common P2P Applications



- Common P2P networks include:
 - G2
 - Bitcoin
 - BitTorrent
 - eDonkey
- Some P2P applications are based on the Gnutella protocol, where each user shares whole files with other users.
- Many P2P applications allow users to share pieces of many files with each other at the same time –this is BitTorrent technology.

How Application Protocols Interact with End-User Applications

Researching Peer-to-Peer File Sharing



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Lab - Researching Peer-to-Peer File Sharing

Objectives

Part 1: Identify P2P Networks, File Sharing Protocols, and Applications

Part 2: Research P2P File Sharing Issues

Part 3: Research P2P Copyright Litigations

Background / Scenario

Peer-to-peer (P2P) computing is a powerful technology that has many uses. P2P networks can be used to share and exchange files, and other electronic materials.

The use of P2P networks to upload, download, or share copyrighted material, such as movies, music, and software, can violate the rights of copyright owners. In the P2P file-sharing context, infringement may occur when one person purchases an authorized copy and then uploads it to a P2P network to share with others. Both the individual who makes the file available and those making copies may be found to have infringed the rights of the copyright owners and may be violating copyright law.

Another problem with P2P file sharing is that very little protection is in place to ensure that the files exchanged in these networks are not malicious. P2P networks are an ideal medium for spreading malware, such as computer viruses, worms, Trojan horses, spyware, adware, and other malicious programs.

In this lab, you will research available P2P file sharing software and identify issues that can arise from the use of this technology.

Required Resources

Device with Internet access

Part 1: Identify P2P Networks, File Sharing Protocols, and Applications

In Part 1, you will research P2P networks and identify popular P2P protocols and applications.

Step 1: Define P2P networking.

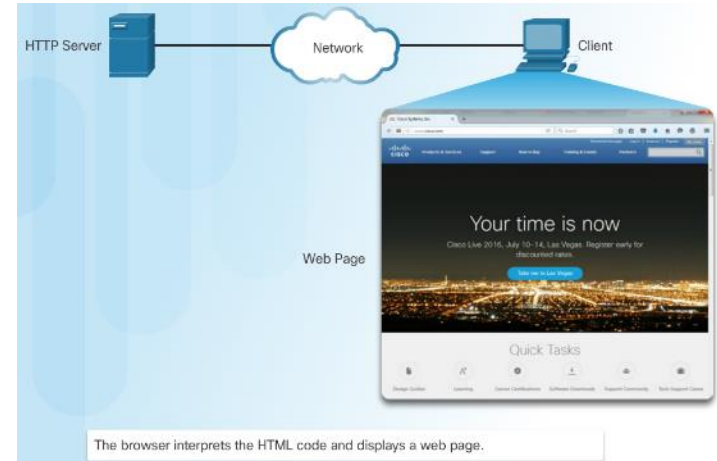
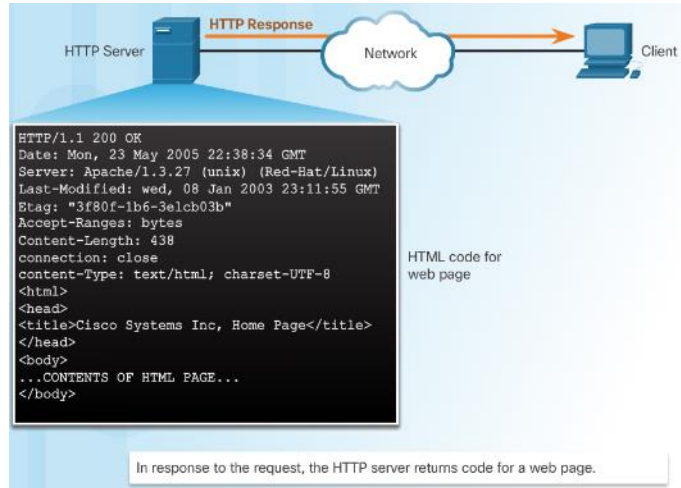
- a. What is a P2P network?

10.2 Well-Known Application Layer Protocols and Services

Hypertext Transfer Protocol and Hypertext Markup Language



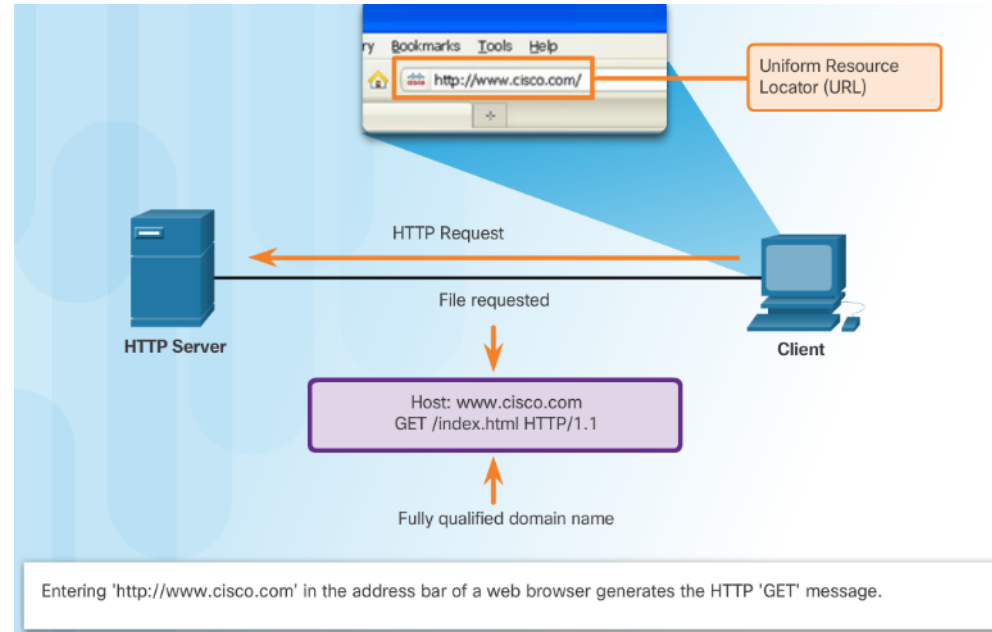
- When a web address or uniform resource locator (URL) is typed into a web browser, the web browser establishes a connection to the web service running on the server, using the HTTP protocol.



Web and Email Protocols

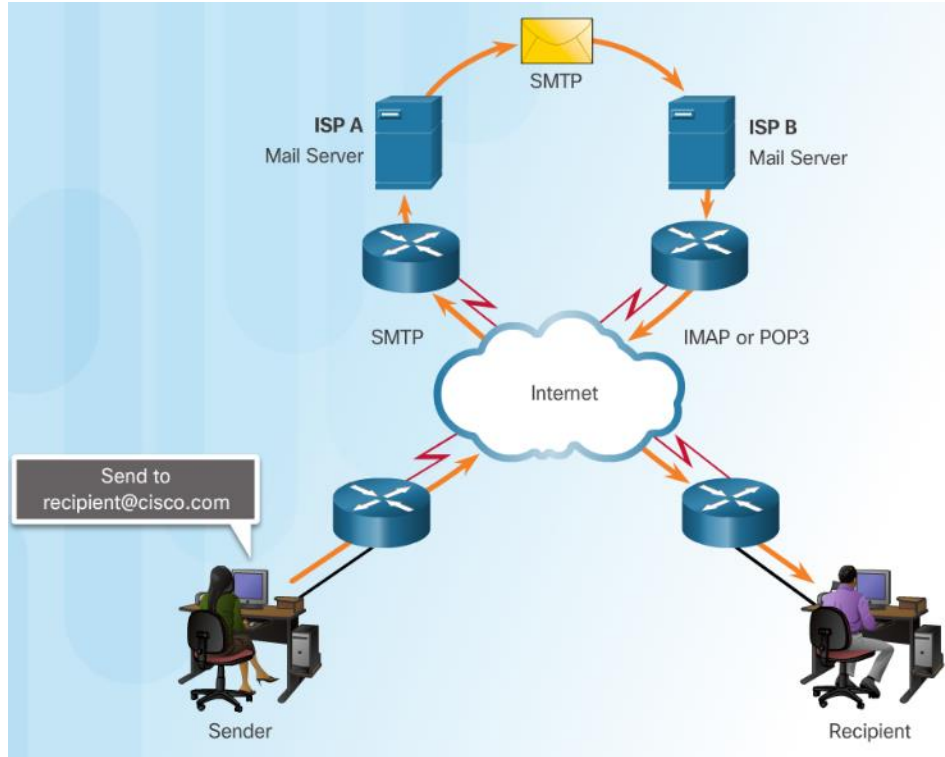
HTTP and HTTPS

- HTTP is a request/response protocol.
- Three common HTTP message types are:
 - GET - A client request for data.
 - POST - Uploads data files to the web server.
 - PUT - Uploads resources or content to the web server.
- HTTP Secure (HTTPS) protocol uses encryption and authentication to secure data.



Web and Email Protocols

Email Protocols

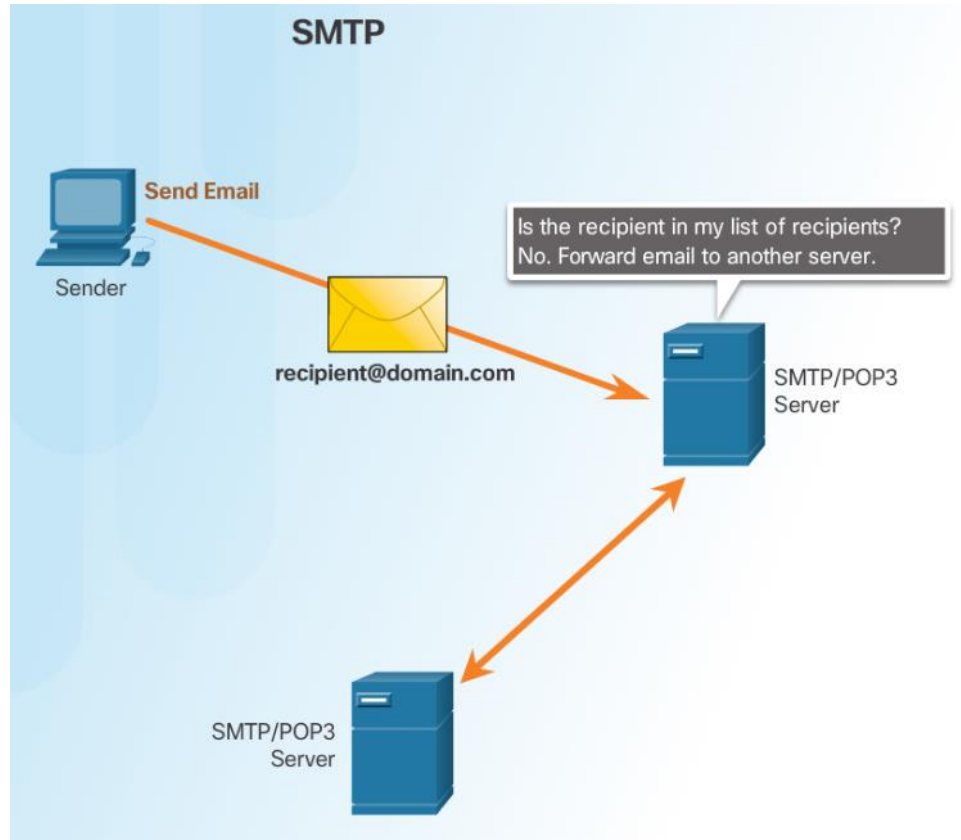


- Email clients communicate with mail servers to send and receive email.
- Mail servers communicate with other mail servers to transport messages from one domain to another.
- Three protocols for email:
 - Simple Mail Transfer Protocol (SMTP) to send email.
 - Post Office Protocol (POP) to retrieve email.
 - Internet Message Access Protocol (IMAP) to retrieve email.

Web and Email Protocols

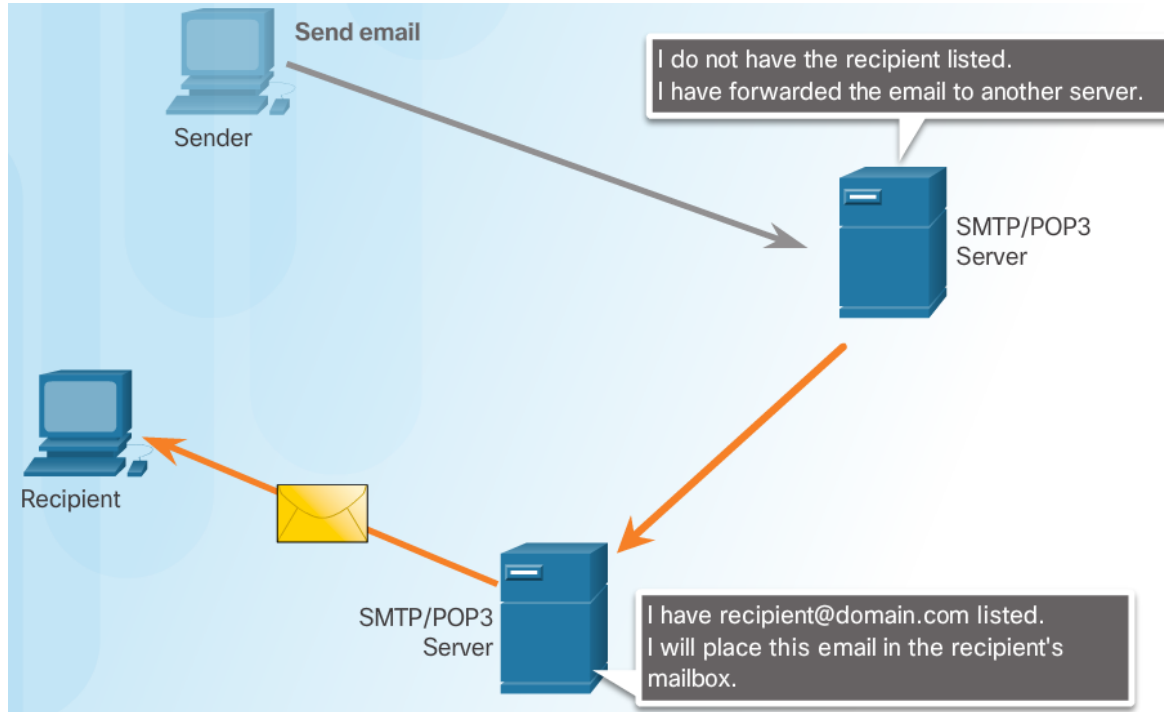
SMTP Operation

- SMTP is used to send email



Web and Email Protocols

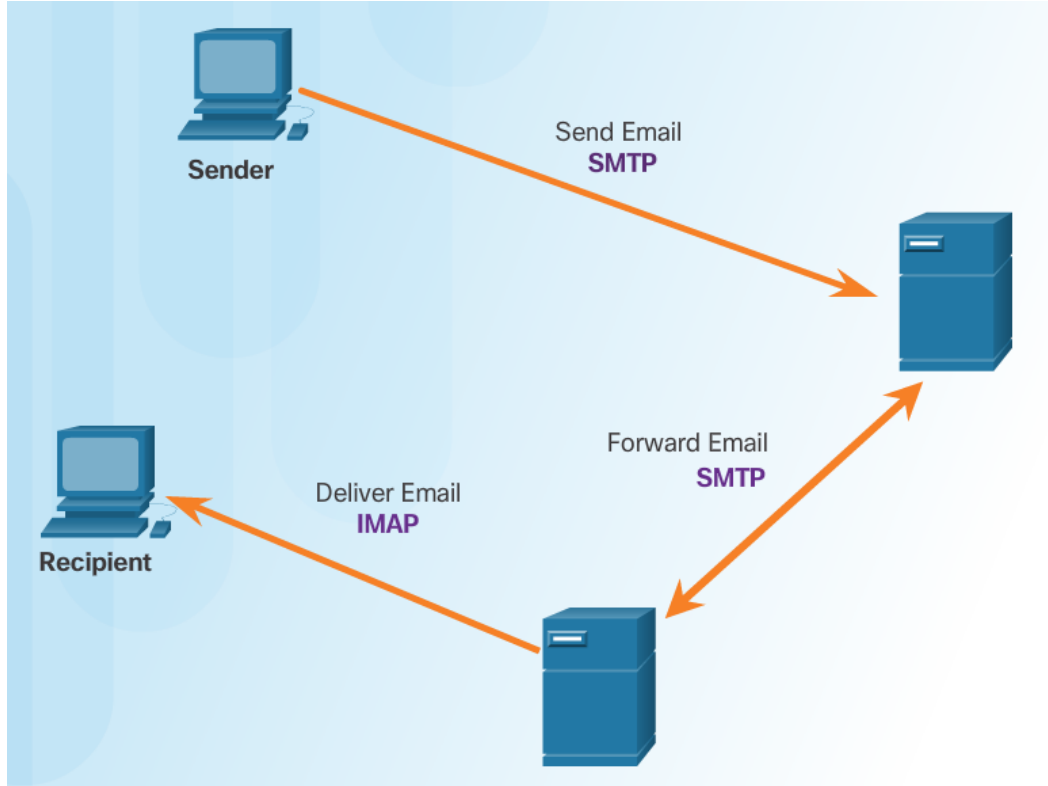
POP Operation



- POP is used to retrieve email from a mail server.
- Email is downloaded from the server to the client and then deleted on the server.

Web and Email Protocols

IMAP Operation



- IMAP is used to retrieve mail from a mail server.
- Copies of messages are downloaded from the server to the client and the original messages are stored on the server.

Web and Email Protocols

Packet Tracer – Web and Email

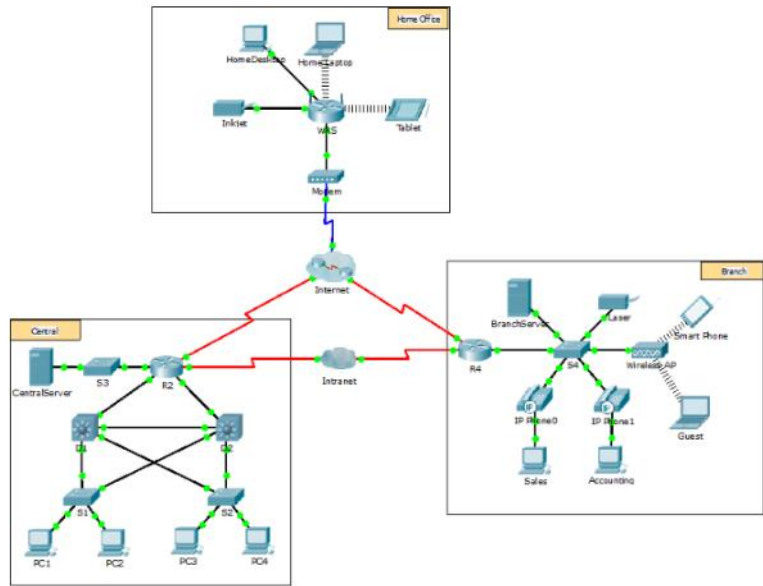


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Packet Tracer - Web and Email

Topology



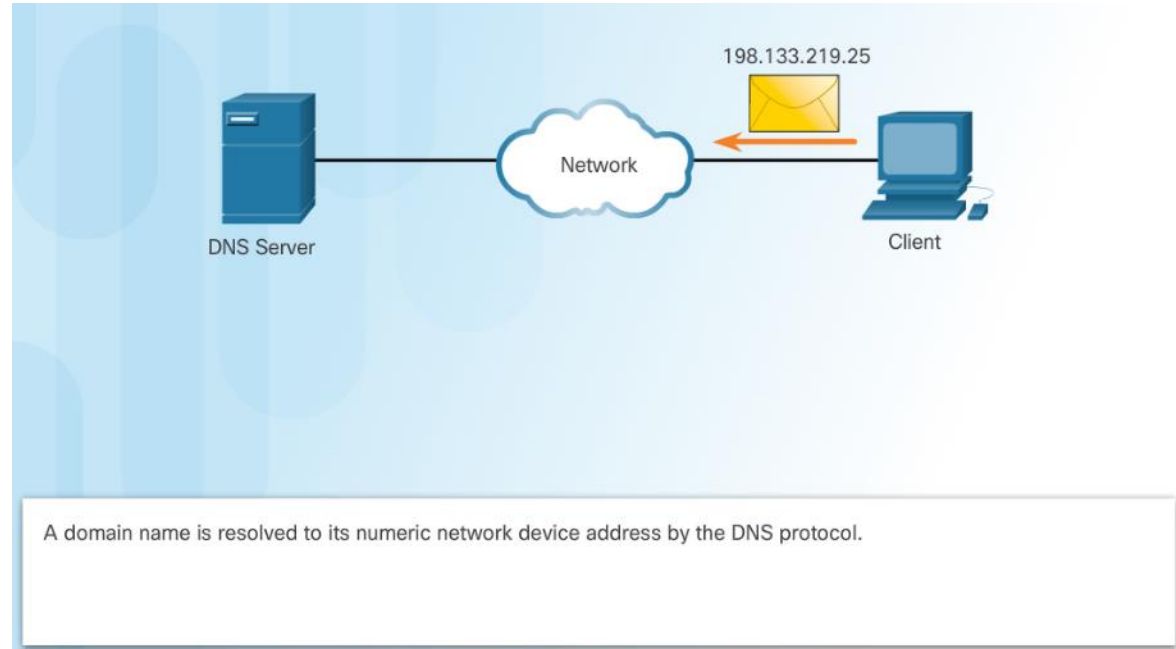
Objectives



IP Addressing Services

Domain Name Service

- Domain names convert the numeric address into a simple, recognizable name.
- The DNS protocol defines an automated service that matches resource names with the required numeric network address.



IP Addressing Services

DNS Message Format

DNS uses the same message format for:

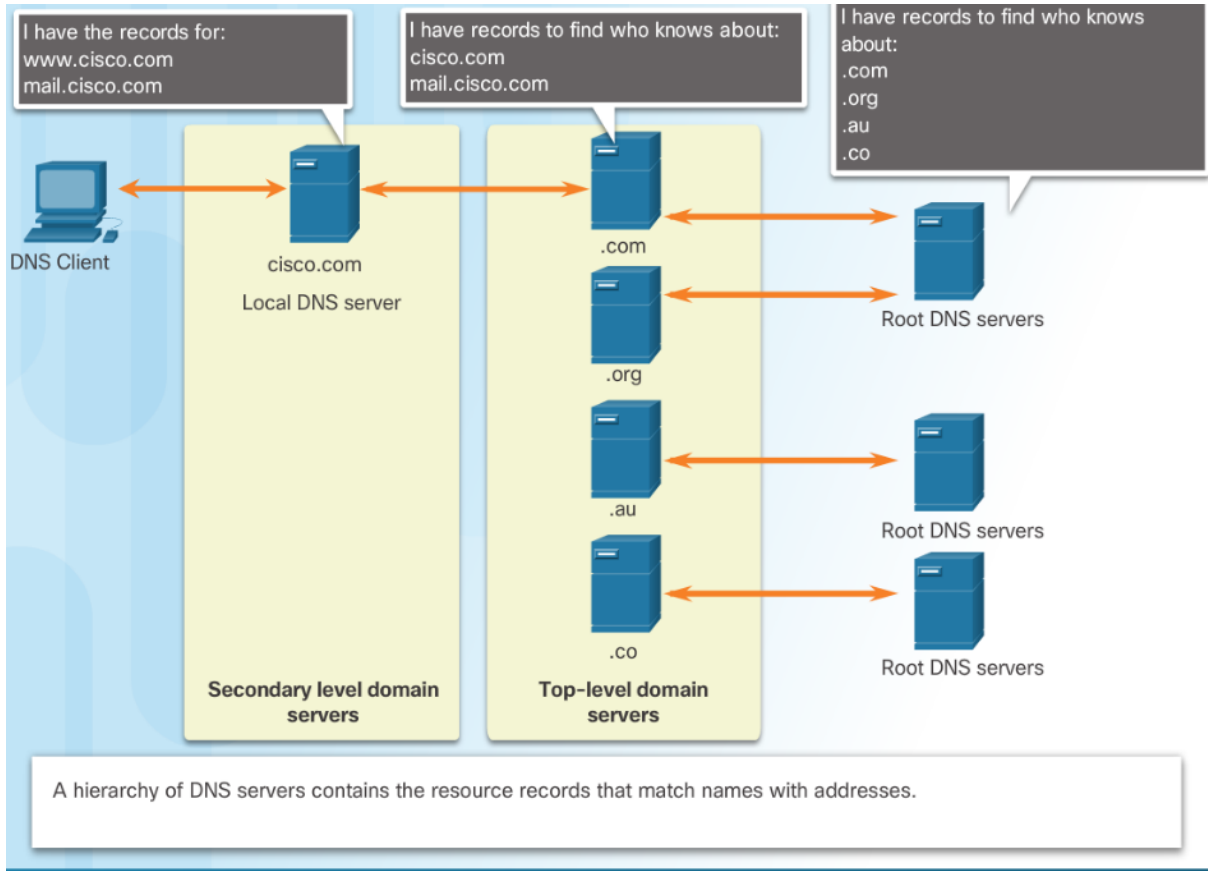
- all types of client queries and server responses
- error messages
- the transfer of resource record information between servers

Header	
Question	The question for the name server
Answer	Resource Records answering the question
Authority	Resource Records pointing toward an authority
Additional	Resource Records holding additional information

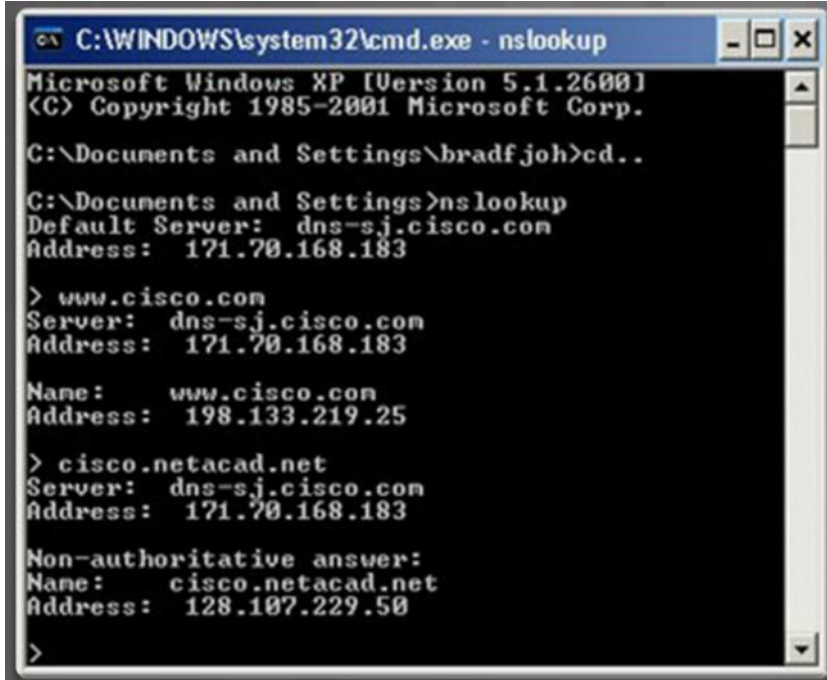
- When a client makes a query, the server's DNS process first looks at its own records to resolve the name.
- If unable to resolve, it contacts other servers to resolve the name.
- The server temporarily stores the numbered address in the event that the same name is requested again.
- The **ipconfig /displaydns** command displays all of the cached DNS entries on a Windows PC.

IP Addressing Services

DNS Hierarchy



The nslookup Command



```
C:\WINDOWS\system32\cmd.exe - nslookup
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\bradfjoh>cd..

C:\Documents and Settings>nslookup
Default Server: dns-sj.cisco.com
Address: 171.70.168.183

> www.cisco.com
Server: dns-sj.cisco.com
Address: 171.70.168.183

Name: www.cisco.com
Address: 198.133.219.25

> cisco.netacad.net
Server: dns-sj.cisco.com
Address: 171.70.168.183

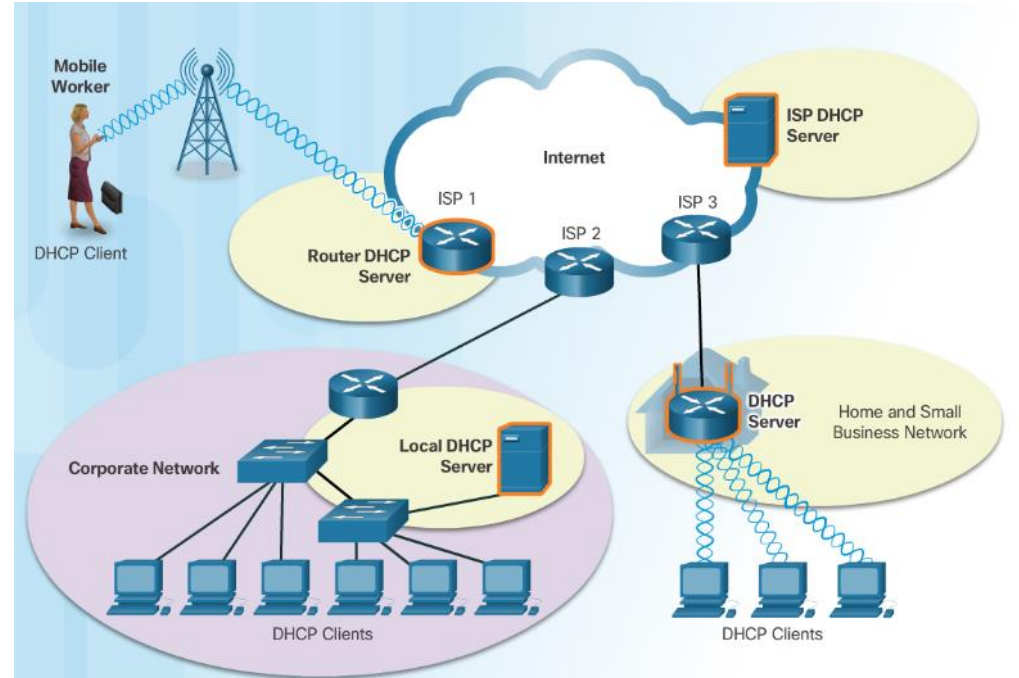
Non-authoritative answer:
Name: cisco.netacad.net
Address: 128.107.229.50

>
```

- **Nslookup** - a utility that allows a user to manually query the name servers to resolve a given host.
- Can also be used to troubleshoot name resolution issues and to verify the current status of the name servers.

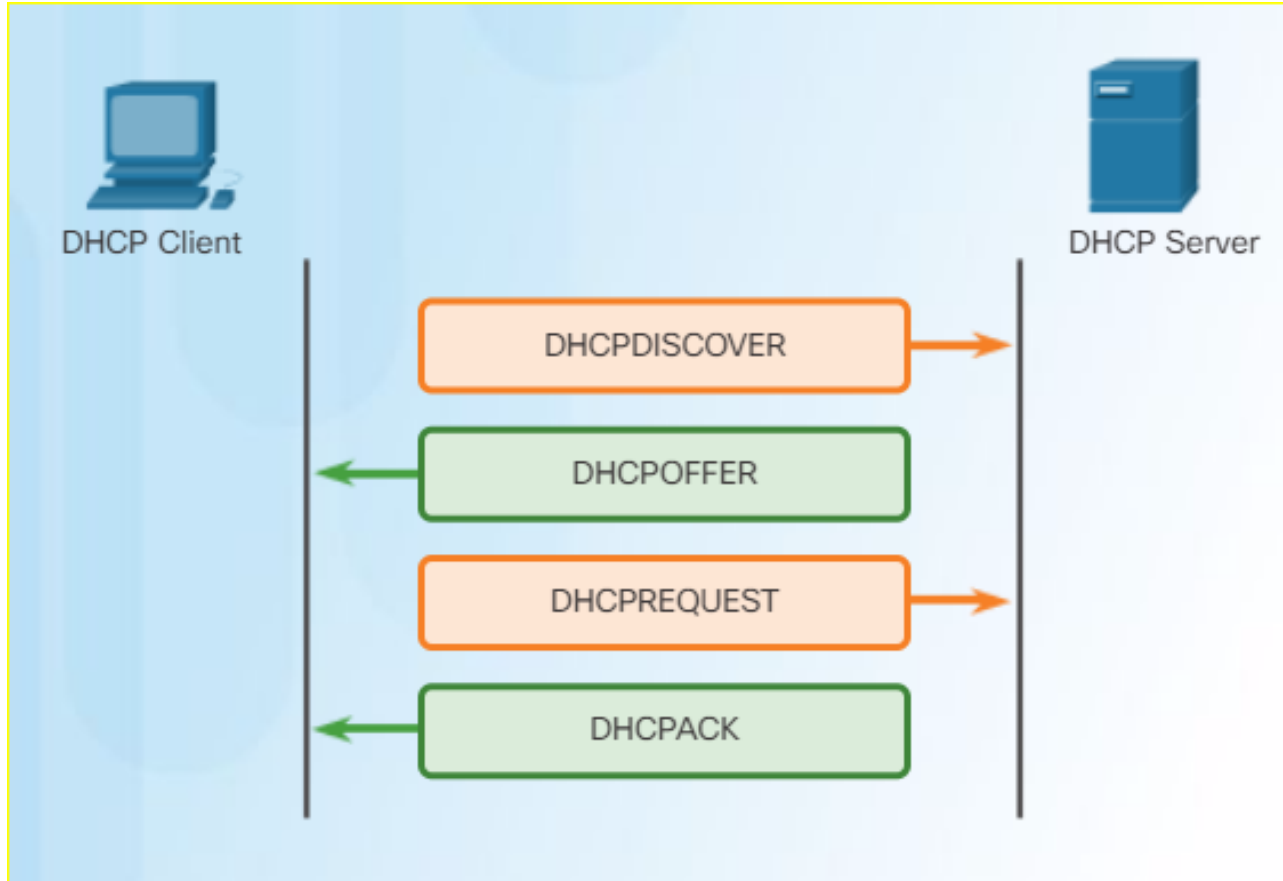
Dynamic Host Configuration Protocol

- The Dynamic Host Configuration Protocol (DHCP) for IPv4 automates the assignment of IPv4 addresses, subnet masks, gateways, and other parameters.
- DHCP-distributed addresses are leased for a set period of time, then returned to pool for reuse.
- DHCP is usually employed for end user devices. Static addressing is used for network devices, such as gateways, switches, and printers.
- DHCPv6 (DHCP for IPv6) provides similar services for IPv6 clients.



IP Addressing Services

DHCP Operation



Packet Tracer – DHCP and DNS Servers

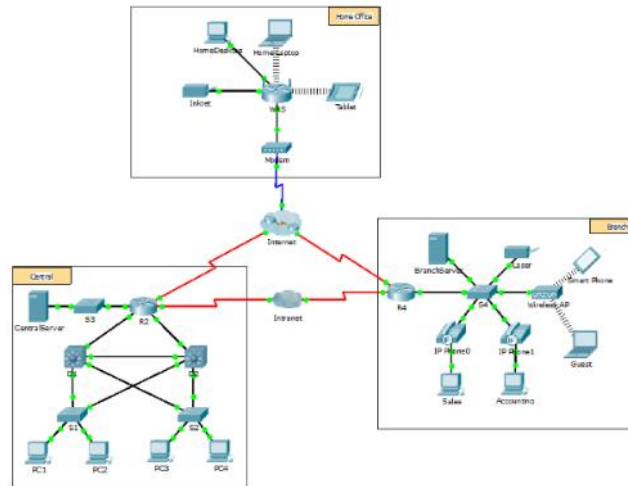


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Packet Tracer - DHCP and DNS Servers

Topology



Objectives

Part 1: Configure Static IPv4 Addressing

Part 2: Configure and Verify DNS Records

Background

In this activity, you will configure and verify static IP addressing and DHCP addressing. You will then

IP Addressing Services

Lab – Observing DNS Resolution



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Lab - Observing DNS Resolution

Objectives

- Part 1: Observe the DNS Conversion of a URL to an IP Address
- Part 2: Observe DNS Lookup Using the nslookup Command on a Web Site
- Part 3: Observe DNS Lookup Using the nslookup Command on Mail Servers

Background / Scenario

The Domain Name System (DNS) is invoked when you type a Uniform Resource Locator (URL), such as <http://www.cisco.com>, into a web browser. The first part of the URL describes which protocol is used. Common protocols are Hypertext Transfer Protocol (HTTP), Hypertext Transfer Protocol over Secure Socket Layer (HTTPS), and File Transfer Protocol (FTP).

DNS uses the second part of the URL, which in this example is www.cisco.com. DNS translates the domain name (www.cisco.com) to an IP address to allow the source host to reach the destination host. In this lab, you will observe DNS in action and use the nslookup (name server lookup) command to obtain additional DNS information. Work with a partner to complete this lab.

Required Resources

- 1 PC (Windows 7 or 8 with Internet and command prompt access)

Part 1: Observe the DNS Conversion of a URL to an IP Address

- Click the **Windows Start** button, type `cmd` into the search field, and press **Enter**. The command prompt window appears.
- At the command prompt, ping the URL for the Internet Corporation for Assigned Names and Numbers (ICANN) at www.icann.org. ICANN coordinates the DNS, IP addresses, top-level domain name system management, and root server system management functions. The computer must translate www.icann.org into an IP address to know where to send the Internet Control Message Protocol (ICMP) packets.

The first line of the output displays www.icann.org converted to an IP address by DNS. You should be able to see the effect of DNS, even if your institution has a firewall that prevents ping, or if the destination server has prevented you from pinging its web server.

Note: If the domain name is resolved to an IPv6 address, use the command `ping -4 www.icann.org` to translate into an IPv4 address if desired.

```
C:\>ping www.icann.org

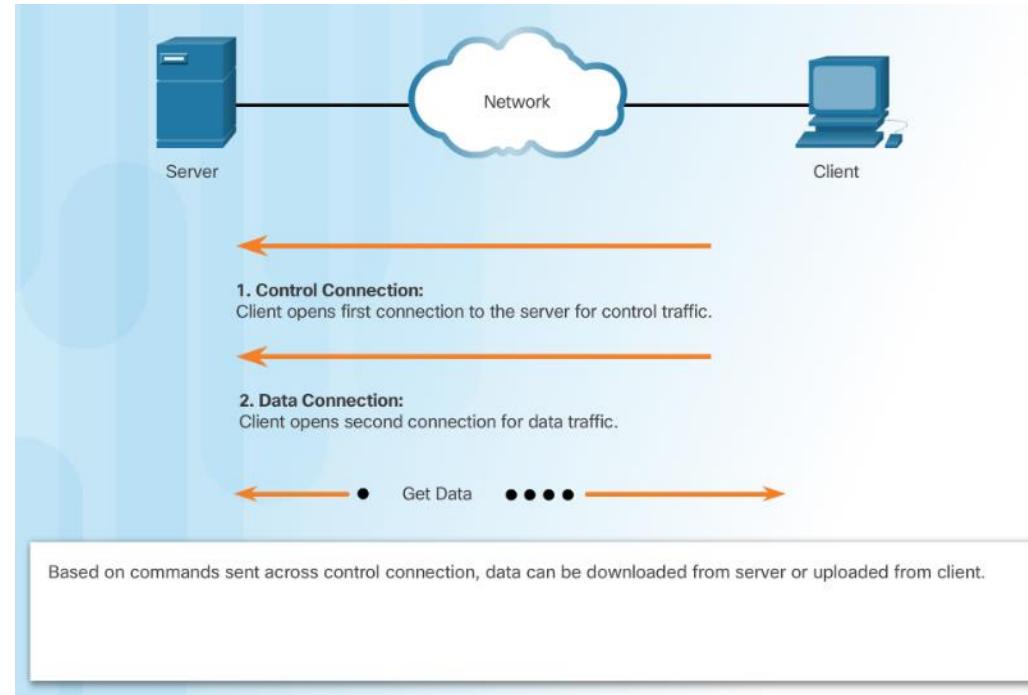
Pinging www.vip.icann.org [192.0.32.71] with 32 bytes of data:
Reply from 192.0.32.71: bytes=32 time=23ms TTL=246
Reply from 192.0.32.71: bytes=32 time=23ms TTL=246
Reply from 192.0.32.71: bytes=32 time=24ms TTL=246
Reply from 192.0.32.71: bytes=32 time=26ms TTL=246

Ping statistics for 192.0.32.71:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 23ms, Maximum = 26ms, Average = 24ms
```

Record the IP address of www.icann.org.

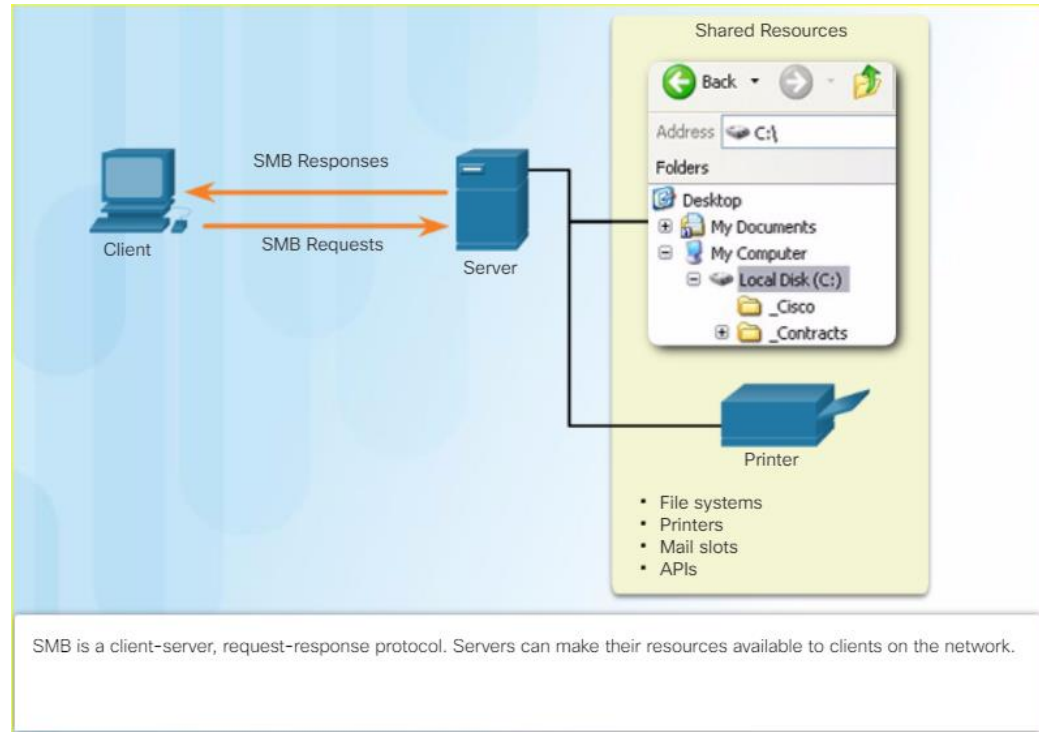
File Transfer Protocol

- FTP requires two connections between the client and the server, one for commands and replies, the other for the actual file transfer:
 - The client establishes the first connection to the server for control traffic using TCP port 21.
 - The client establishes the second connection to the server for the actual data transfer using TCP port 20.



Server Message Block

- The Server Message Block (SMB) is a client/server file sharing protocol:
 - SMB file-sharing and print services have become the mainstay of Microsoft networking.
 - Clients establish a long-term connection to servers and can access the resources on the server as if the resource is local to the client host.



File Sharing Services

Packet Tracer - FTP

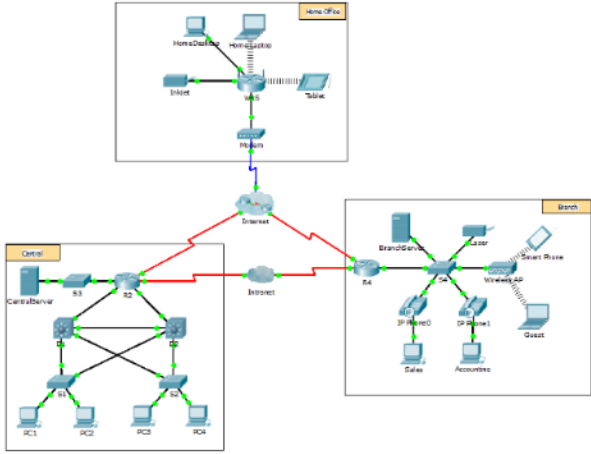


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Packet Tracer - FTP Servers

Topology



Objectives

- Part 1: Configure FTP Services on Servers
- Part 2: Upload a File to the FTP Server
- Part 3: Download a File from the FTP Server

Background

In this activity, you will configure FTP services. You will then use the FTP services to transfer files between clients and the server.

File Sharing Services

Lab – Exploring FTP



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Lab - Exploring FTP

Objectives

- Part 1: Use FTP from a Command Prompt
- Part 2: Use FTP in a Browser
- Part 3: Download an FTP File Using WS_FTP LE (Optional)

Background / Scenario

The File Transfer Protocol (FTP) is part of the TCP/IP suite. FTP is used to transfer files from one network device to another network device. Windows includes an FTP client application that you can execute from the command prompt. There are also free graphical user interface (GUI) versions of FTP that you can download. The GUI versions are easier to use than typing from a command prompt. FTP is frequently used for the transfer of files that may be too large to send using email.

When using FTP, one computer is normally the server and the other computer is the client. When accessing the server from the client, you need to provide a username and password. Some FTP servers have a user named **anonymous**. You can access these types of sites by simply typing "anonymous" for the user, without a password. Usually, the site administrator has files that can be copied but does not allow files to be posted with the anonymous user. Furthermore, FTP is not a secure protocol because the data is not encrypted during transmission.

In this lab, you will learn how to use anonymous FTP from the Windows command-line C:\> prompt. You will access an anonymous FTP server using your browser. Finally, you will use the GUI-based FTP program, WS_FTP LE.

Required Resources

- 1 PC (Windows 7 or 8 with access to the command prompt, Internet access, and WS_FTP LE installed (optional))

Part 1: Use FTP from a Command Prompt

- Click the **Windows Start** button, type **cmd** in the search field, and press **Enter** to open a command window.
- At the C:\> prompt type **ftp ftp.cdc.gov**. At the prompt that says **User (ftp.cdc.gov:(none)):** type **anonymous**. For the password, do not type anything. Press **Enter** to be logged in as an anonymous user.

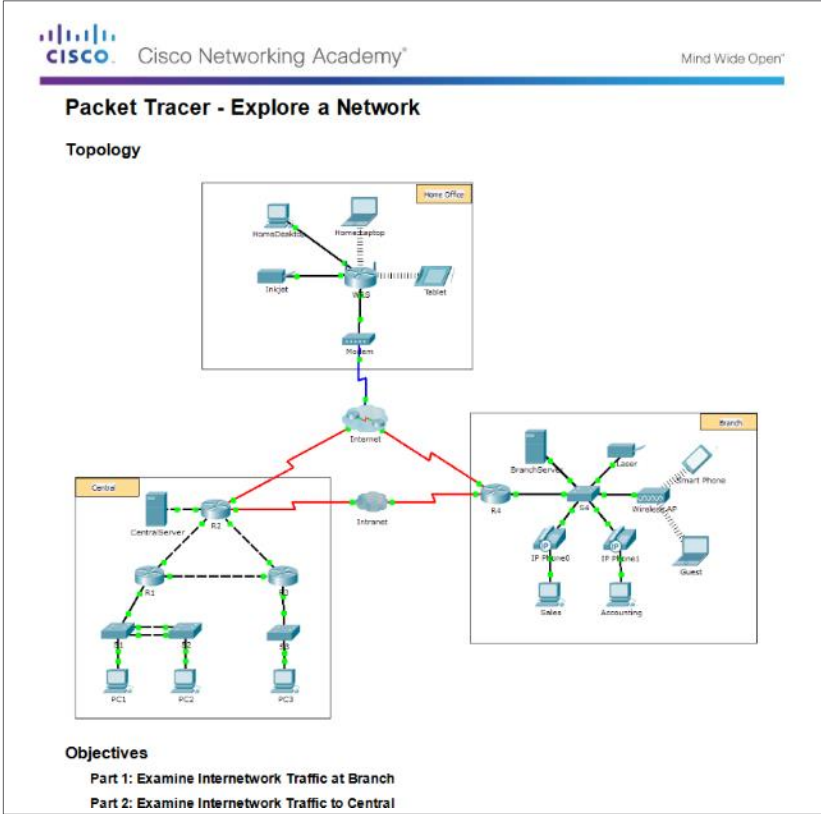
```
Microsoft Windows [Version 6.1.7600]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\Nigel>ftp ftp.cdc.gov
Connected to ftp.cdc.gov.
220 Microsoft FTP Service
User (ftp.cdc.gov:(none)): anonymous
031 Anonymous access allowed, send identity (e-mail name) as password.
Password:
030 Anonymous user logged in.
ftp>
```

10.3 Summary

Conclusion

Packet Tracer - Explore a Network



Conclusion

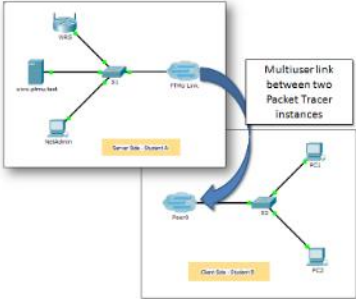
Packet Tracer - Multiuser - Tutorial

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Packet Tracer Multiuser - Tutorial

Topology



Addressing Table

Device	IP Address	Subnet Mask	DNS Server
www.ptmu.test	10.10.10.1	255.0.0.0	10.10.10.1
PC	10.10.10.10	255.0.0.0	10.10.10.1

Objectives

Part 1: Establish a Local Multiuser Connection to another Instance of Packet Tracer

Part 2: Verify Connectivity across a Local Multiuser Connection

Background

The multiuser feature in Packet Tracer allows multiple point-to-point connections between multiple instances of Packet Tracer. This first Packet Tracer Multiuser (PTMU) activity is a quick tutorial demonstrating the steps to establish and verify a multiuser connection to another instance of Packet Tracer within the same LAN. Ideally, this activity is meant for two students. However, it can also be completed as a solo activity simply by opening the two separate files to create two separate instances of Packet Tracer on your local machine.

Conclusion

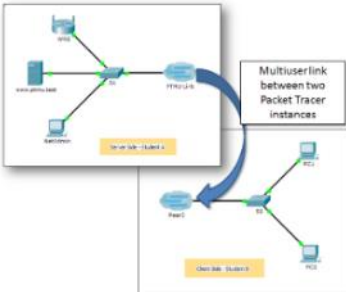
Packet Tracer Multiuser - Implement Services

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Packet Tracer Multiuser - Implement Services

Topology



Addressing Table

Device	IP Address	Subnet Mask
Server Side Player		
WRS	172.16.1.254	255.255.255.0
S1	172.16.1.1	255.255.255.0
www.ptmu.test	172.16.1.5	255.255.255.0
NetAdmin	DHCP Assigned	DHCP Assigned
Client Side Player		
S2	172.16.1.2	255.255.255.0
PC1	DHCP Assigned	DHCP Assigned
PC2	DHCP Assigned	DHCP Assigned

Chapter 10: Application Layer

- Explain the operation of the application layer in providing support to end-user applications.
- Explain how well-known TCP/IP application layer protocols operate.

